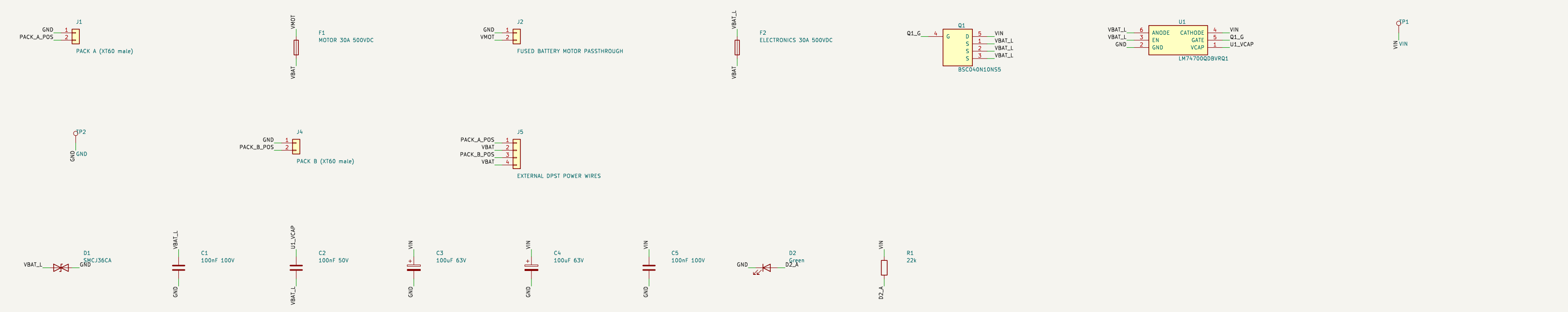
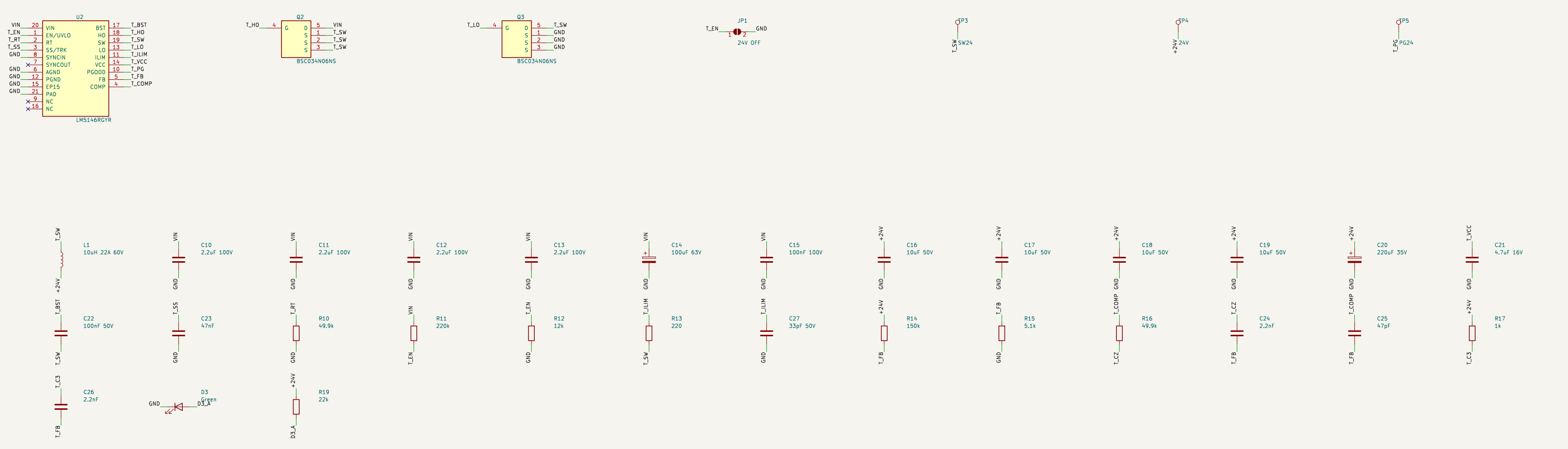
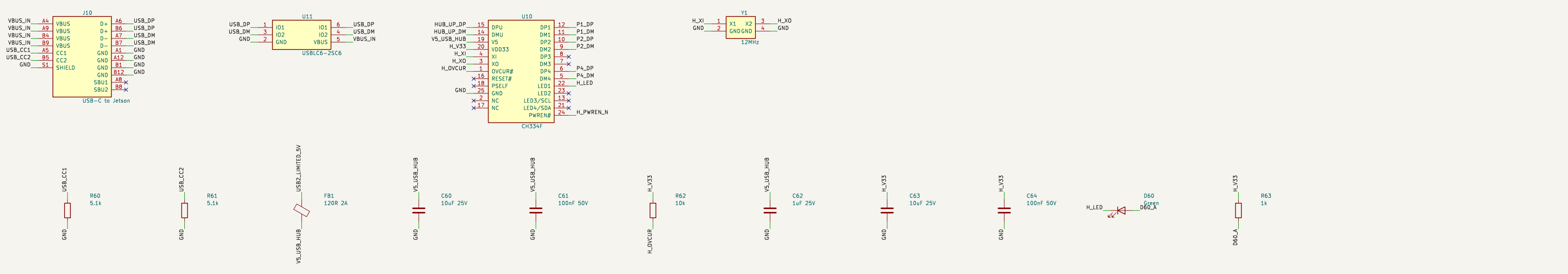
BASE. A. Battery input, motor pass-through, logic protection



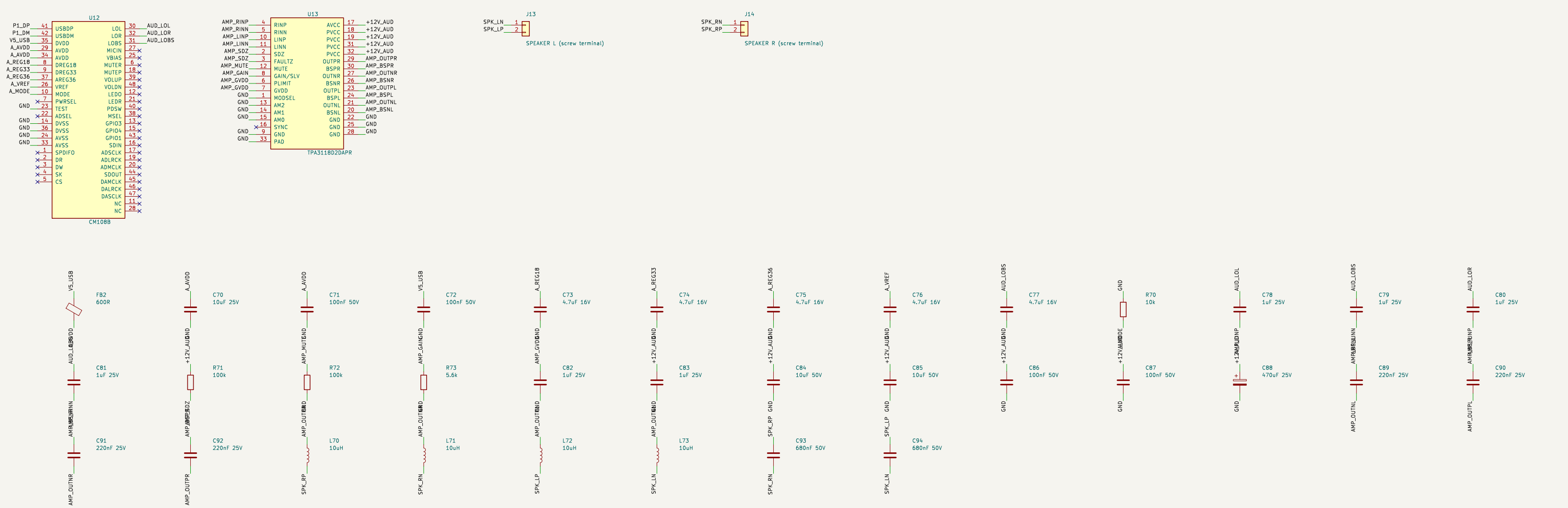
BASE. B. 24V for Thor dev kit (8A connector ceiling; unqualified) (LM5146 sync buck, 200 kHz)



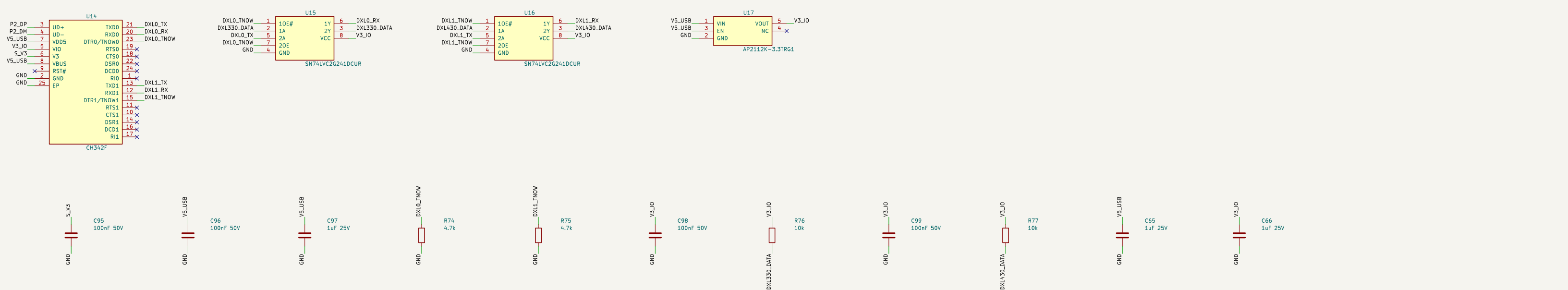
BASE. I. USB in + 4-port hub (CH334F)



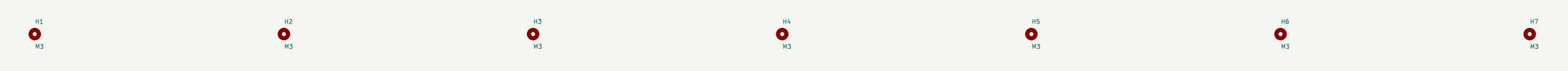
BASE. J. USB audio (CM108B) + 2x15W class-D amp (TPA3118D2) on the 12V audio rail



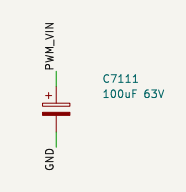
BASE. K. Dynamixel USB interface: CH342F dual serial + half-duplex buffers (U2D2 replacement)



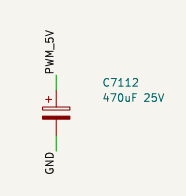
BASE. H. Mounting



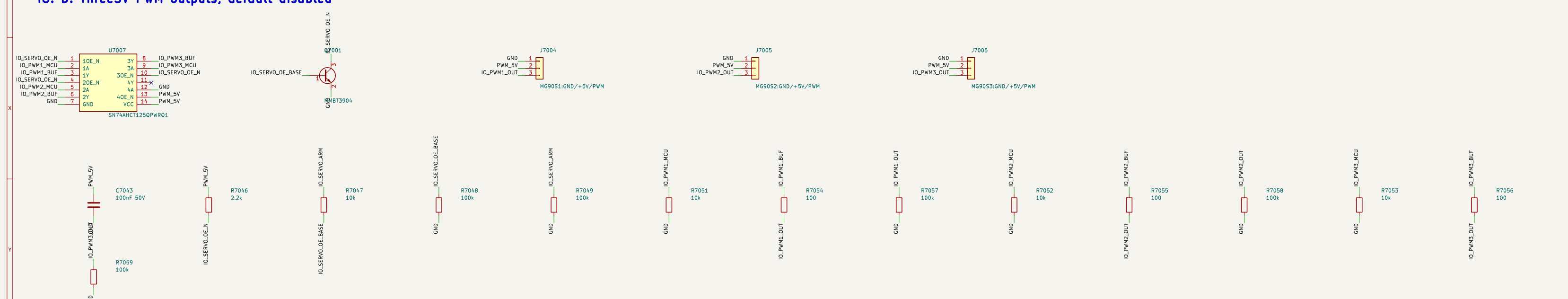
IO. A. Input bulk



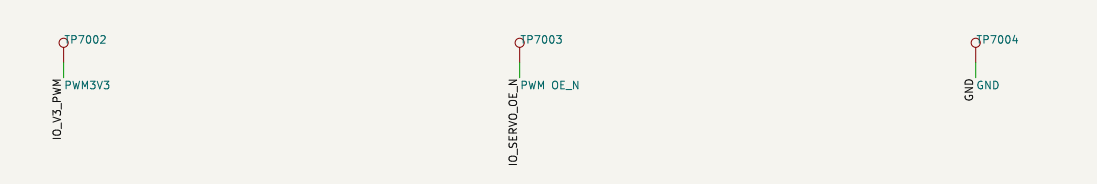
IO. A. Servo pulse bulk



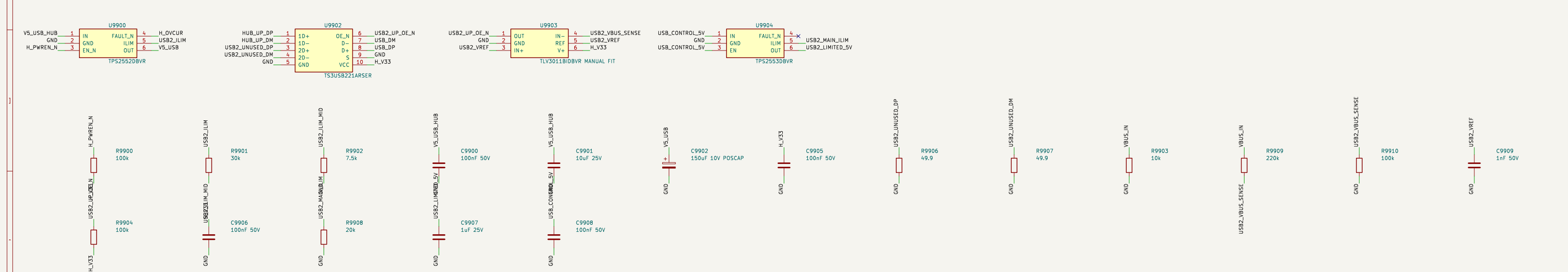
IO. D. Three5V PWM outputs, default disabled



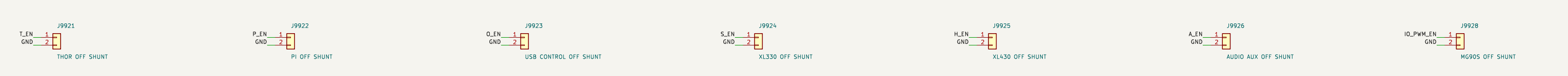
IO. E. Service



USB2. USB2 CONTROL. Local supply and host-controlled data attachment

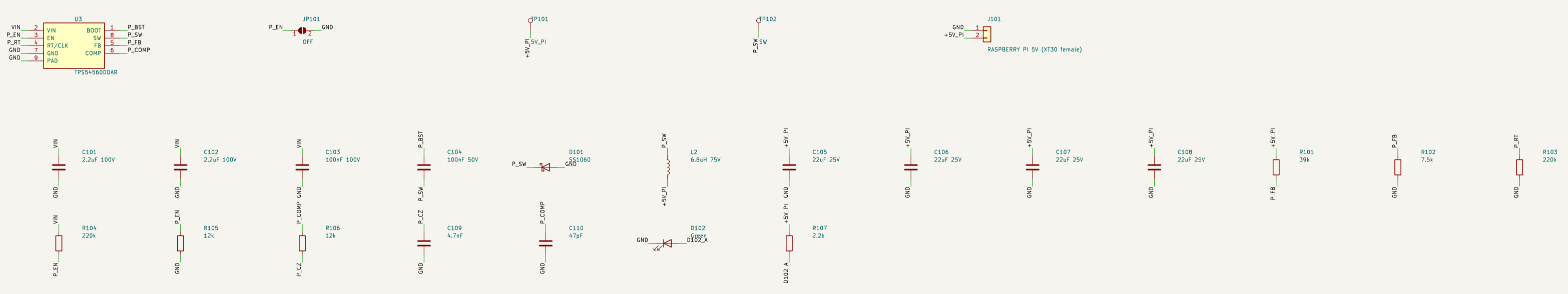


SERVICE. Optional local regulator OFF shunts

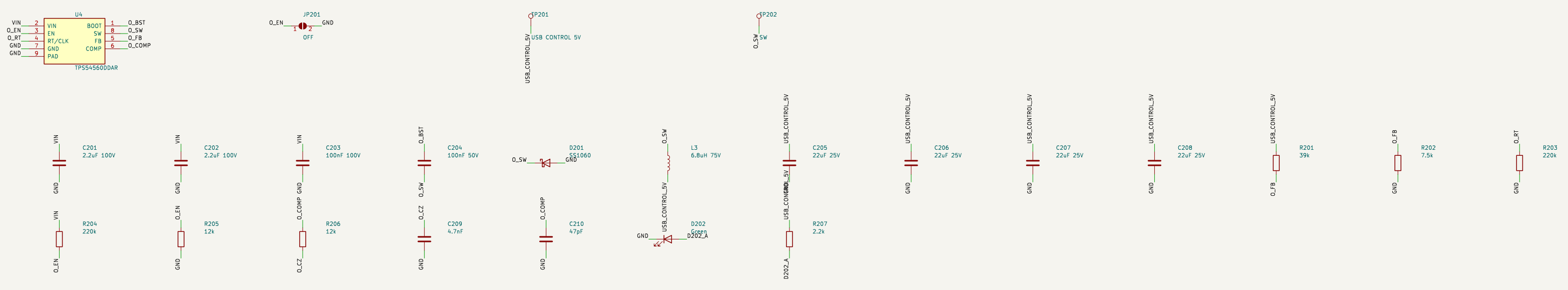


T_EN
GND
THOR OFF SHUNT
P_EN
GND
PI OFF SHUNT
O_EN
GND
USB CONTROL OFF SHUNT
S_EN
GND
XL330 OFF SHUNT
H_EN
GND
XL430 OFF SHUNT
A_EN
GND
AUDIO AUX OFF SHUNT
IO_PWM_EN
GND
MG90S OFF SHUNT

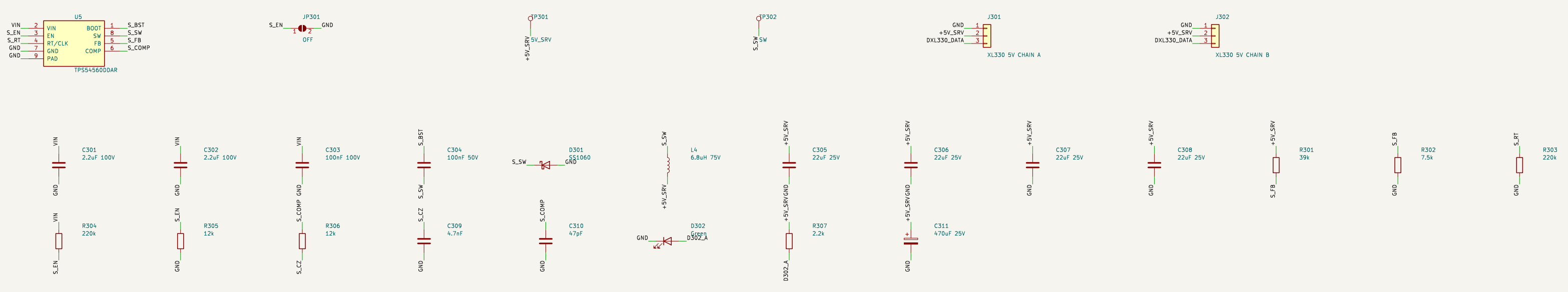
BASE. C. 5V / 5A Raspberry Pi



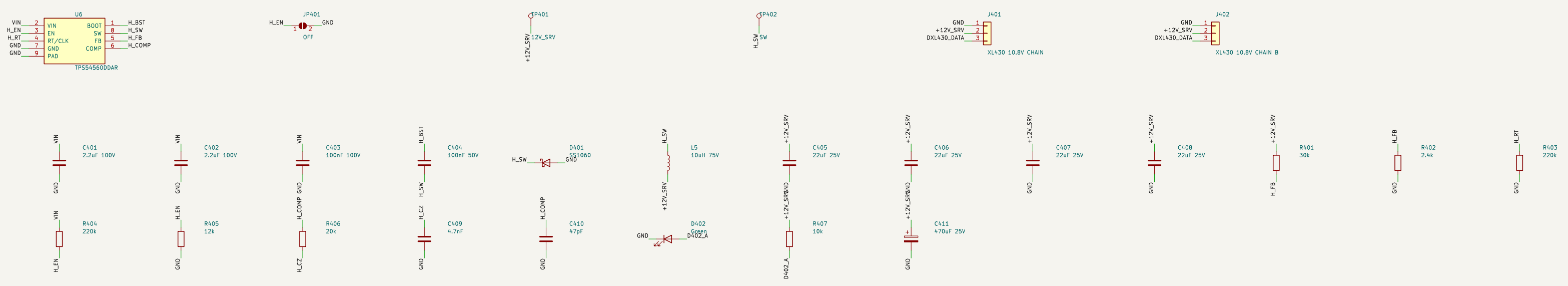
BASE. D. 5V / 5A OAK-D depth camera



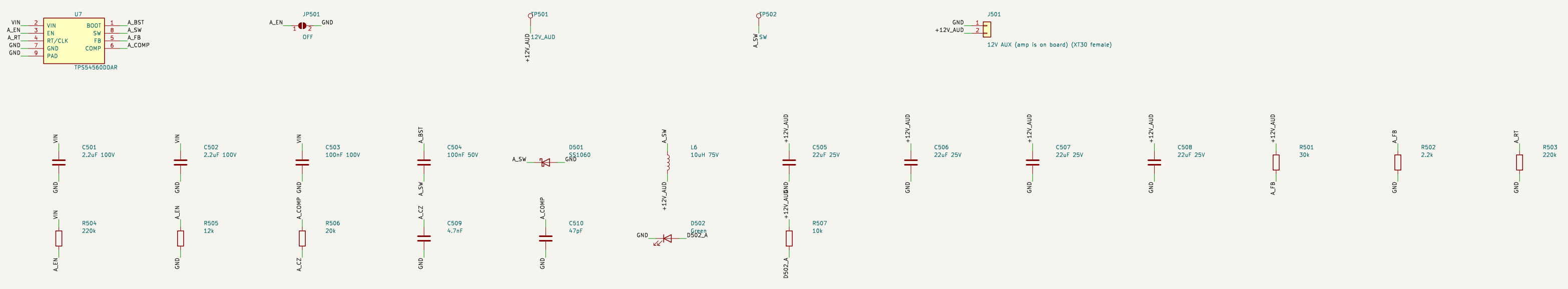
BASE. E. XL330 servo power, 5 V, two daisy-chain sockets



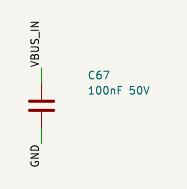
BASE. F. XL430 servo power, 10.8 V, one daisy-chain socket



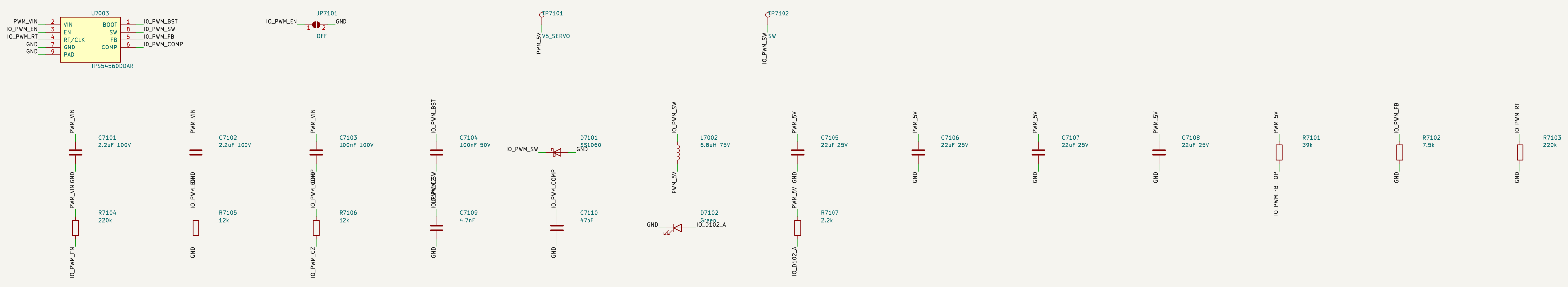
BASE. G. 11.7V / 5A Audio amplifier



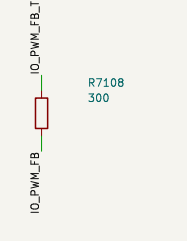
BASE. I. USB attachment slew control



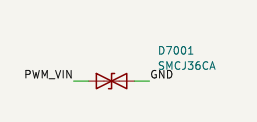
IO. A. Dedicated MG90S servo5V (5A target unqualified)



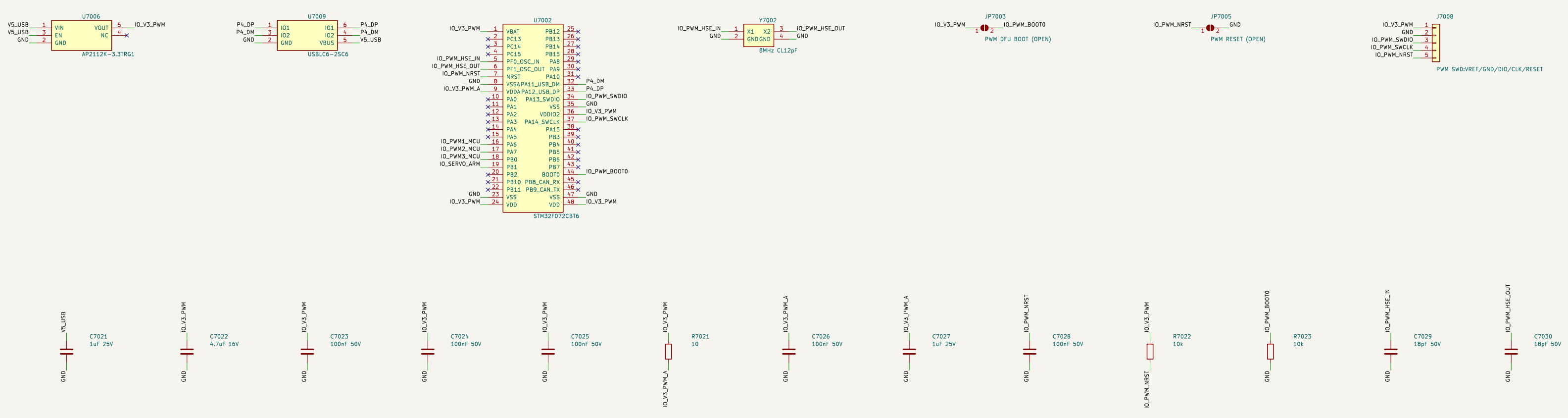
IO. A. Servo5V trim to4.992V nominal



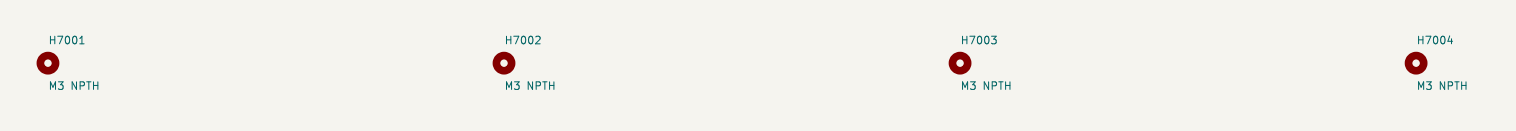
IO. A. Local input surge footprint



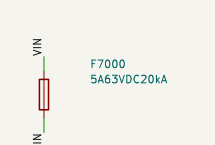
IO. B2. PWM USB MCU and independent USB logic supply



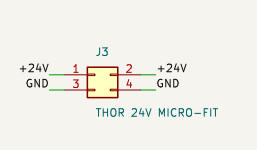
IO. E. Mounting



BRANCH. PWM converter input fuse



THOR OUTPUT. Standard four-contact power cable



THOR 24V MICRO-FIT

MOUNT. Harness support

H9910
M3 NYLON SUPPORT
H9911
M3 NYLON SUPPORT
H9912
M3 NYLON SUPPORT

ENGINEERING PROTOTYPE; final layout/assembly review and bench qualification required. External mechanical DPST disconnect: STOP ROBOT BEFORE OFF. 27-33V intended battery operation,33.6V full; individual pack-source fuses required. J2 is original fused BATTERY-VOLTAGE motor passthrough to existing Flipsky controllers. No motor converter, brake clamp, external brake resistor, thermal contact or onboard CAN. Existing motor controllers retain their previous control connection and braking configuration. USB2 VBUS is sense only; local5V powers audio/UART/PWM through the CH334 hub. Hub port3 unused; no configuration EEPROM. OAK-D/microphone connect externally to Jetson USB3. Seven local shunts shortENtoGND=OFF; they do not i

8S battery, 33.6V full. Engineering review only. See docs/HANDOFF.md.

Sheet: /
File: robot_pdb.kicad_sch

Title: Robot power distribution board

Size: User	Date:	**Rev: 0.11 compact layout**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1